Corpus-based automatic construction of topic maps by self-organizing maps

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Abstract. Topic maps standard (ISO-13250) have been gradually recognized as an emerging standard for information exploration and knowledge organization in the web era. One advantage of topic maps is that they enable a user to navigate and access the documents he wants in an organized manner, rather than browsing through hyperlinks that are generally unstructured and often misleading. Nowadays, the topic maps are generally manually constructed by domain experts or users since the functionality and feasibility of automatically generated topic maps still remains unclear. In this work we proposed a text mining scheme for automatic topic maps construction. We applied a machine learning algorithm on a corpus of information resources to identify the topics and discover the relations among these topics. Necessary components of topic maps such as topics, topic types, topic occurrences, and topic associations may be automatically revealed by our method.

1 Introduction

Nowadays the users of the WWW try to access the huge amount of documents on the web (or specifically, the web pages) by querying through a search engine or browsing through hyperlinks within a web page. However, modern search engines generally reveal too many or too few web pages in responding to a user's query simply because we were not able to correctly and effectively represent the web pages as well as the queries. Efforts on effective indexing and searching of web pages [1] may help to some extent. However, most of the works devoted to syntactic analysis of web pages, rather than discovering their semantic contents. Therefore, the notation of relevancy in such works was defined syntactically but not semantically. On the other hand, browsing through the hyperlinks within a web page may also be misleading because most of the hyperlinks were constructed manually by the authors. Although it is sure that the authors should know the semantic meaning of their pages and make the hyperlinked pages being semantically relevant, browsing through hyperlinks may still fail to fulfill the users' needs due to the following reasons. First, a user often has to browse through several pages to reach his destination because he has no idea about where to start and, as a result, he is forced to start from the beginning, i.e.

those home pages that contain the most general idea that he intends to find. Besides, in most cases he also has to find the relevant home pages in a try-and-error manner. Seconds, the authors of the pages may not refer to all pages in the corpus during the authoring process; thus limits the coverage of the relevancy. As a result, a user has to browse lots of pages to collect a bunch of relevant pages that may be useful to him. Accompany with other obvious deficiencies of traditional searching and browsing schemes, development of methods for more efficient and effective information exploration on the web have attracted many researchers and organizations. Topic map standard [4] is one of the results of such efforts.

Topic maps provide a general, powerful, and user-oriented way to navigating the information resources under consideration in any specified domain. A topic map provides a uniform framework that not only identifies important subjects from an entity of information resources and specifies the information resources that are semantically related to a subject, but also explores the relations among these subjects all in a uniform framework. When a user needs to find some specific information on a pool of information resources, he could just peek into the topic maps of this pool, select the topic he thought interesting, and the topic maps will show him the information resources that are related to this topic as well as its related topics. He will also recognize the relationships among these topics and the roles the topics play in such relationships. With the help of the topic maps, we no longer have to browse through a set of hyperlinked documents and hope they may eventually reach the information we need in a finite amount of time even when we know nothing about where should we start. We also don't have to squeeze some words from our mind and hope that they may perfectly symbolize the idea we interest and are conceived well by a search engine to obtain reasonable result. Topic maps provide us a way to navigate and organize information, as well as to create and maintain knowledge in an infoglut.

To construct a topic map for a set of information resources, human intervention is unavoidable in present time. We need human effort in tasks such as selecting topics, identifying their occurrences, and revealing their associations. Such situation is considered acceptable when the topic maps are used for navigation purpose and the volume of the information resource is considerably small. However, a topic map should not only be a 'topic navigation map' [1]. Besides, the volume of information resource under consideration is generally large enough to prevent manual construction of topic maps. To expand the applicability of topic maps, some kind of automatic process should involve during the construction of topic maps. The degree of automation in such construction process may differ for different users with different needs: one may only need a friendly interface to automate the topic map authoring process, and one may try to automatically identify every components of a topic map for a set of information resources from the ground up. In this work, we recognize the importance of topic maps not only as a navigation tool but also an ideal scheme for knowledge acquisition and representation. According to such recognition, we tried to develop a scheme based

[1] This is the name previously used before 1996.

on a proposed text mining approach to automatically construct topic maps for a set of information resource. Our approach is opposite to the navigation task performed by a topic map to obtain information. We extracted knowledge from a corpus of documents in a specific domain to construct a topic map. Both the power and limitation of such novel act were discussed in this work.

2 Related Work

Topic map standard is an emerging standard so there were few works about it. Most of the early works about topic maps focused on providing introductory materials [8, 2]. Few of them were devoted to automatic construction of topic maps. Two works that addressed this issue were reported in [9] and [7]. Rath [9] discussed a framework for automatic generation of topic maps according to a so-called 'topic map template' and a set of generation rules. The structural information of topics is maintained in the template. They used a generator to interpret the generation rules and extract necessary information that fulfills the template to create the topic map. However, both the rules and the template were to be constructed explicitly and probably manually. Moore [7] discussed topic map authoring and how software may support it. He argued that the automatic generation of topic maps is a useful first step in the construction of a production topic map. However, the real value of a topic map comes through the involvement of people in the process. This argument is true if the knowledge that contained in the topic maps can only be obtained by human efforts. Fully automatic generation process is possible only when such knowledge may be discovered from the underlying set of information resources through an automated process, which is generally known as knowledge discovery from texts or *text mining*.

3 A Brief Review of Topic Maps

In this section we only provide a brief review of topic maps and focus on the components that construct a topic map. Details about the topic maps were provides in [4, 8, 2]. A topic map is constructed by a set of *topics* and the relationships among these topics. A topic is an identified subject about the underlying information resources and is an aggregate of topic properties such as topic names, topic types, topic occurrences, and association types, etc. A topic usually has names that we can easily refer to the topic. A topic name consists of a base name and two optional names, namely display name and sort name. For example, "PAKDD02", "PAKDD 2002", and "pakdd02" may be the base name, display name, and sort name respectively for a topic in the topic map of all documents related to PAKDD-2002. We will use the base name to refer to a topic in the context. A topic is an instance of zero or more topic types. For example, PAKDD02 is a topic of type "conference" and Taipei is a topic of types "city" and "conference location". A topic may be linked to one or multiple information resources that are relevant to the topic in some way. Such resources are called occurrences of the topic. For example, the web page with

URL "`http://arbor.ee.ntu.edu.tw/pakdd02/`" is an occurrence of the topic PAKDD02. The file `pakdd02-cfp.pdf` is also an occurrence of the same topic. An occurrence may be of any number of different types, which are called the roles of the occurrence. An example of the role of occurrence `pakdd02-cfp.pdf` mentioned above is "document".

The relationships between topics are described by the topic associations. A topic association is a link element that asserts a relationship between two or more topics. Examples of topic associations are "PAKDD 2002 is *held in* Taipei", "Taipei *is in* Taiwan", and "Harry Potter is *written by* J. K. Rowling". Associations may also have types that are called association types. For example, the types of the associations in above example are "*held-in*", "*is-in*", and "*written-by*", respectively. Besides, each topic that participates in an association has a corresponding association role which states the role played by the topic in the association. For example, in the association "Harry Potter is written by J. K. Rowling" the roles of the two topics "Harry Potter" and "J. K. Rowling" are "book" and "author", respectively.

A topic map is composed by a set of topics and the relationships between the topics in a specific domain. A schematic example of a topic map is depicted in Fig. 1. In Fig. 1 a solid line indicates a association between two topics and a dotted line indicates an occurrence of a topic. Topics with different types are enclosed by different diagrams. Such distinction is also applied to those occurrences with different roles. Notice that we only associate a topic with one type for illustration purpose. A topic may have more than one types in a standard topic map.

4 The Text Mining Process

Before we can create topic maps, we should first perform a text mining process, which is similar to the work described in [6] and [11], on the set of information resources to reveal the relationships among the information resources. Here we only consider those information resources that can be represented in regular texts. Examples of such resources are web pages, ordinary books, technical specifications, manuals...etc. The set of information resources is collectively known as *the corpus* and individual resource is referred as a document in the following text. To reveal the relationships between documents, the popular self-organizing map (SOM) [5] algorithm was applied to the corpus to cluster documents. We adopted the vector space model [10] to transform each document in the corpus into a binary vector. These document vectors were used as input to train the map. We then applied two kinds of labeling process to the trained map and obtained two feature maps, namely the document cluster map (DCM) and the word cluster map (WCM). In the document cluster map each neuron represents a document cluster that contains several similar documents with high word co-occurrence. In the word cluster map each neuron represents a cluster of words that reveal the general concept of the corresponding document cluster associated

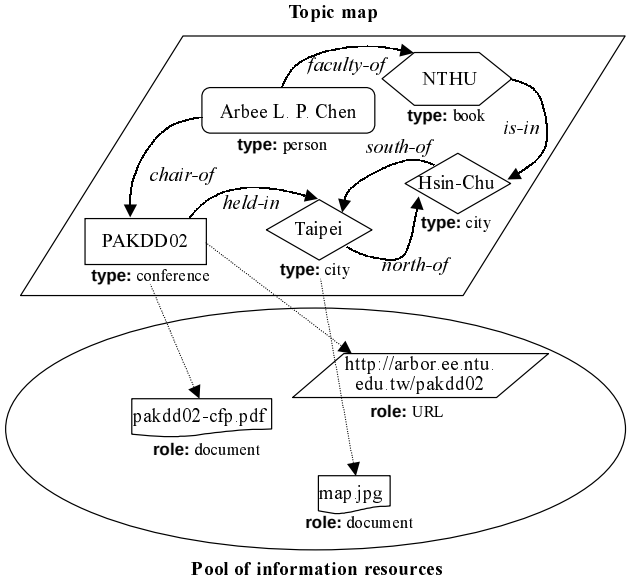


Fig. 1. A schematic example of a topic map

to the same neuron in the document cluster map. The detail training process is described in [6].

Since the DCM and the WCM use the same neuron map, a neuron represents a document cluster as well as a word cluster simultaneously. By linking the DCM and the WCM according to the neuron locations we may discover the underlying ideas of a set of related documents. This is essentially a text mining process because we can discover related terms among a set of related documents through their co-occurrence patterns that are hard to extract directly. By virtue of the SOM algorithm, terms that are often co-occurred will tend to be labeled to the same neuron or neighboring neurons because the neurons are willing to learn all of them simultaneously. Thus the co-occurrence patterns of indexed terms may be revealed. Moreover, the terms that associated to the same neuron also reveal the common themes of the associated documents of the neuron. Essentially these related terms construct a thesaurus that is derived from the context of the underlying documents rather than their grammatical meaning. Thus the indexed terms associated to the same neuron in the WCM compose a pseudo-document which represents the general concept of the documents associated to that neuron. Therefore our approach may also generate a thesaurus automatically, another text mining application.

The text mining process described above provides us a way to reveal the relationships between the topics of the documents. We introduce here a method to identify topics and the relationships between topics. The method also arrange these topics in a hierarchical manner according to their relationships. As we mentioned before, a neuron in the DCM represents a cluster of documents that

contain words that often co-occurred in these documents. Besides, documents that associate with neighboring neurons contain similar set of words. Thus we may construct a super-cluster by combining neighboring neurons. To form a super-cluster, we first define the distance between two clusters:

$$D(i,j) = H(||\mathbf{G}_i - \mathbf{G}_j||), \tag{1}$$

where i and j are the neuron indices of the two clusters and $\mathbf{G}_i$ is the two-dimensional grid location of neuron i. $||\mathbf{G}_i - \mathbf{G}_j||$ measures the Euclidean distance between the two coordinates $\mathbf{G}_i$ and $\mathbf{G}_j$. $H(x)$ is a bell-shaped function that has maximum value when $x = 0$. We also defined the dissimilarity between two clusters:

$$\mathcal{D}(i,j) = ||\mathbf{w}_i - \mathbf{w}_j||, \tag{2}$$

where $\mathbf{w}_i$ denotes the synaptic weight vector of neuron i. We may then compute the *supporting cluster similarity* $\mathcal{S}_i$ for a neuron i from its neighboring neurons by

$$\begin{aligned} S(i,j) &= \frac{\mathrm{doc}(i)\mathrm{doc}(j)}{F(D(i,j)\mathcal{D}(i,j))} \\ \mathcal{S}_i &= \sum_{j \in B_i} S(i,j), \end{aligned} \tag{3}$$

where $\mathrm{doc}(i)$ is the number of documents associated to neuron i in the document cluster map and B_i is the set of neuron index in the neighborhood of neuron i. The function $F : \mathbf{R}^+ \to \mathbf{R}^+$ is a monotonically increasing function. A *dominating neuron* is the neuron which has locally maximal supporting cluster similarity. We may select dominating neuron by the following algorithm:

1. Find the neuron with the largest supporting cluster similarity. Selecting this neuron as dominating neuron.
2. Eliminate its neighbor neurons so that they will not be considered as dominating neurons.
3. If there is no neuron left or the number of dominating neurons exceeds a predetermined value, stop. Otherwise go to step 1.

A dominating neuron may be considered as the centroid of a super-cluster, which contains several clusters. We assigned every cluster to a super-cluster by the following method. The ith cluster (neuron) is assigned to the kth super-cluster if

$$\mathcal{D}(i,k) = \min_l \mathcal{D}(i,l), \qquad l \text{ is a super}-\text{cluster}. \tag{4}$$

A super-cluster may be thought as a category that contains several sub-categories. Let C_k denote the set of neurons that belong to the kth super-cluster, or category. The category topics are selected from those words that associated with these neurons in the WCM. For all neurons $j \in C_k$, we selected the n^*th word as the category topic if

$$\sum_{j \in C_k} w_{j_{n^*}} = \max_{1 \le n \le N} \sum_{j \in C_k} w_{j_n}. \tag{5}$$

Eq. 5 selects the word that is the most important to a super-cluster since the components of the synaptic weight vector of a neuron reflects the willingness that the neuron wants to learn the corresponding input data, i.e. words.

The topics that are selected by Eq. 5 form the top layer of the category hierarchy. To find the descendants of these topics in the hierarchy, we may apply the above process to each super-cluster and obtain a set of sub-categories. These sub-categories form the new super-clusters that are on the second layer of the hierarchy. The category structure can then be revealed by recursively applying the same category generation process to each new-found super-cluster. We decreased the size of neighborhood in selecting dominating neurons when we tried to find the sub-categories.

5 Automatic Topic Maps Construction

The text mining process described in Sec. 4 reveals the relationships between documents and words. Furthermore, it may identify the topics in a set of documents, reveals the relationships among the topics, and arrange the topics in a hierarchical form. The result of such text mining process can be used to construct topic maps. We will discuss the different steps in topic maps construction in the following subsections.

5.1 Identifying Topics and Topic Types

The topics in the constructed topic map can be selected as the topics identified by Eq. 5. All identified topics in every layer of the hierarchy can be used as topics. Since topics in different layers of the hierarchy represent different levels of coverage, we may constrain the coverage of topics in the topic map by limiting the depth of hierarchy that we select topics from. If we only used topics in higher layers, the number of topics is small but the topics represent more general topics. The coverage level can be set explicitly in the beginning of the construction process or determined dynamically during the construction process. A way to determine the number of topics or coverage level is by the size of the self-organizing map.

The topic types can also be determined by the constructed hierarchy. As we mentioned before, a topic on higher layers of the hierarchy represents a more general concept than those on lower layers. For a parent-child relationship between two concepts on two adjacent layers, the parent topic should represent a general concept of its child topic. Therefore, we may use the parent topic as the type of its child topics. Such usage also fulfills the requirement of the topic map standard that a topic type is also a topic.

5.2 Identifying Topic Occurrences

The occurrences of a identified topic are easy to obtain after the text mining process. Since a topic is a word labeled to a neuron in the WCM, its occurrences

can be assigned as the documents labeled to the same neuron in the DCM. That is, let a topic τ be labeled to neuron A in the WCM, the occurrences of τ should be those documents labeled to the same neuron A in the DCM. For example, if the topic 'text mining' was labeled to the 20th neuron in the WCM, all documents labeled to the 20th neuron in the DCM should be the occurrences of this topic. Furthermore, we may create more occurrences of this topic by allowing documents labeled to lower levels of the hierarchy being also included. For example, if neuron 20 in the above example located on the second level of a topic hierarchy, we may also allow the clusters of documents associated with topics below this level being also occurrences of this topic. Another approach is to use the DCM directly such that we also include the documents associated with the neighboring neurons of the neuron associated with a topic as its occurrences.

5.3 Identifying Topic Associations

The associations among topics can be identified by two ways in our method. The first is to use the developed hierarchy structure among topics. A topic is associated with the other if there existed a path between them. We should limit the lengths of such paths to avoid establishing associations between pairs of unrelated topics. For example, if we limited the length to 1, only topics that are direct parents and children are associated with the topic under consideration. The type of such associations is essentially an instance-class association. The second way to identify topic associations simply examines the WCM and finds the associations. To establish associations to a topic τ, we first find the neuron A that τ is labeled to. We then established associations between τ and every topic associated with some neighboring neuron of A. The neighboring neurons are selected from a neighborhood of A that are arbitrarily set by the creator. Obviously, a large neighborhood will create many associations. We should at least create associations between τ and other topics associated to the same neuron A since they are considered as very related topics in the text mining process. The association types are not easy to reveal by the method since we did not fully reveal the semantic relations among neurons after text mining process. An alternative method to determine the association type between two topics is to use the semantic relation defined in a well-developed ontology such as WordNet [3].

6 Experimental Result

We performed the experiments on a set of news articles that were collected from a newswire site [2]. Two corpora were constructed in our experiments. The first corpus (CORPUS-1) contains 100 news articles posted in Aug. 1-3, 1996. The second corpus (CORPUS-2) contains 3268 documents posted during Oct. 1-9, 1996. A word extraction process was applied to the corpora to extract Chinese words. Total 1475 and 10937 words were extracted from CORPUS-1 and

[2] `http://www.cna.com.tw`

CORPUS-2, respectively. To reduce the dimensionality of the feature vectors we discarded those words that occur only once in a document. We also discarded the words that appear in a manually constructed stoplist and reduced the number of words to 563 and 1976 for CORPUS-1 and CORPUS-2, respectively. A reduction rate of 62% and 82% is achieved for the two corpora, respectively.

To train CORPUS-1, we constructed a self-organizing map that contains 64 neurons in 8×8 grid format. The number of neurons was determined experimentally such that a better clustering can be achieved. Each neuron in the map contains 563 synapses. The initial training gain was set to 0.4 and the maximal training time was set to 100. These settings were also determined experimentally. We tried different gain values ranged from 0.1 to 1.0 and various training time setting ranged from 50 to 200. We simply adopted the setting which achieved the most satisfying result. After training we labeled the map by documents and words respectively and obtained the DCM and the WCM for CORPUS-1. The above process was also applied to CORPUS-2 with a 20×20 neuron map and obtained the DCM and the WCM for CORPUS-2.

We applied the hierarchy generation process after the clustering process to the DCM to obtain the category hierarchies. In our experiments we limited the number of dominating neurons to 10 and the depths of hierarchy to 2 and 3 for CORPUS-1 and CORPUS-2, respectively. The category topics of super-clusters were also identified. In Figure 2 we show the overall hierarchies developed from CORPUS-1. Each tree depicts a hierarchy where the number on the root node depicts the super-cluster found in the first iteration of the generation process. Let STAGE-n denote the nth iteration of the hierarchy generation process. The number of hierarchies is the same as the number of super-clusters found at STAGE-1. Each leaf node in a tree represents a cluster in the DCM. The parent node of some child nodes in level n of a tree represents a super-cluster found in STAGE-$(n-1)$. For example, the root node of the largest tree in Figure 2 has a number 35, specifying that neuron 35 is one of the ten dominating neurons found in STAGE-1. This node has 10 children that are the 10 dominating neurons obtained in STAGE-2. These child nodes comprise the second level of the hierarchy. The third level nodes are obtained after STAGE-3. The number enclosed in a leaf node is the neuron index of its associated cluster in the DCM. The identified category topics are used to label every node in the hierarchies. In Figure 3 we only show the largest hierarchy developed from CORPUS-2 due to space limitation.

The topic maps of these two corpora can then be generated based on the above results. We show part of the overall topic map for CORPUS-1 in Fig. 4. In the figure we used the same two-layer view of topic maps that separate the topics and their occurrence as in Fig. 1. Topics were translated to English for reference. The occurrences of a topic were collectively shown in the same box. We have used the subjects of articles to refer to the articles. A subject starts with a '@' in the figure.

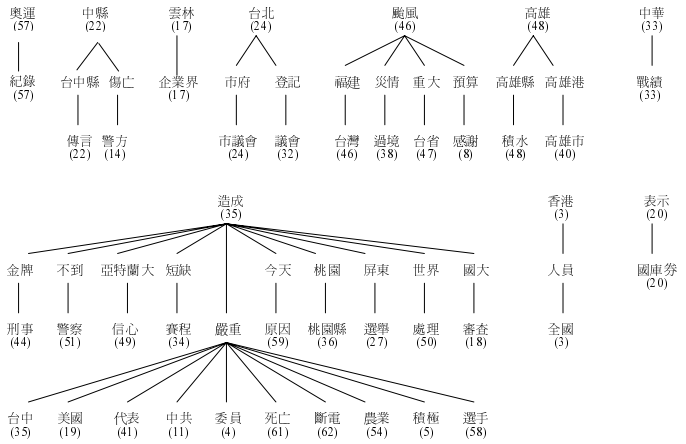


Fig. 2. The category hierarchies of CORPUS-1

7 Conclusions and Discussions

In this work we presented a novel approach for automatic topic map construction. The approach started from applying a text mining process on a set of information resources. Two feature maps, namely the document cluster map and the word cluster map, were created after the text mining process. We then applied a category hierarchy development process to reveal the hierarchical structure of the document clusters. Some topics were also identified by such process to indicate the general subjects of those clusters located in the hierarchy. We may then create topic maps according to the two maps and the developed hierarchy automatically. So far, our method is the only approach that may create topic maps full automatically.

We provide some discussions of the method here. Our method is based on the result of a text mining process that may discover knowledge from a set of text documents. This requires that the set of information resources being represented in textual form. However, this is not quite a constraint since almost all information resources on the web nowadays are stored in textual form. Non-textual information resources such as images, videos, sounds, etc. may require either a multimedia data mining technique or a alternate textual representation scheme to fit in our method.

The need of an automatic construction method such as ours should increase when the size of corpus becomes larger. Although we may, or even should, need human intervention in the construction process, full automatic process is useful in several aspects. First, the selection of topics reflects the general view about the information resources of the author. Different authors may have various views and therefore create different topic maps that may confuse a user. Users need to select one of the topic maps and hope it will eventually lead them to their goals. However, if it fails after several navigation steps, most of the users will turn

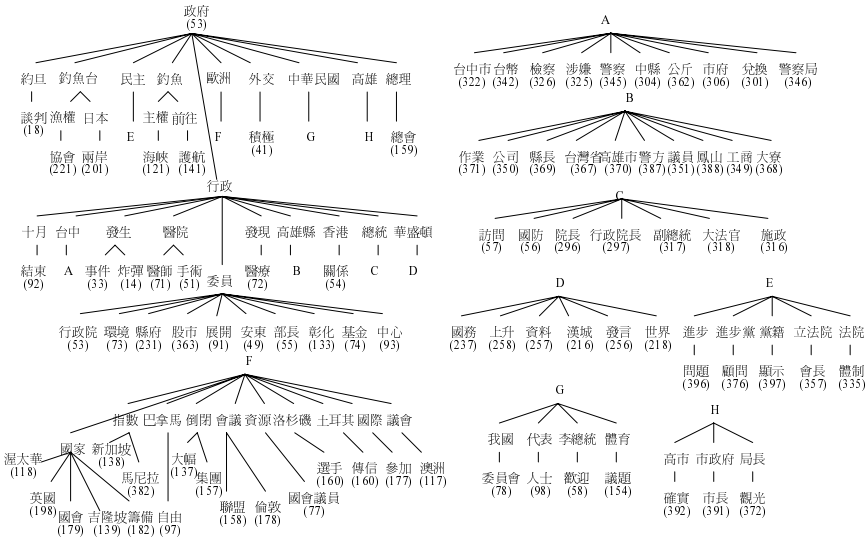


Fig. 3. One of the category hierarchies developed from CORPUS-2

to another topic map and try it again. This may waste considerable time. The problem is it is unlikely that a human being may understand the whole set of information resources thoroughly when the size of the set is considerably large. Some sort of support is generally necessary to acquire knowledge from the corpus of information. On the other hand, a fully automatically constructed topic map may not produce the best topic map that fulfill every users' need. However, a good topic map should capture the necessary semantic information of the set of information resources. We consider this as a guideline in constructing topic maps and found the text mining process may help. Through our approach, there are even no need for domain experts to create topic maps. Second, our method may provide a possible way to evaluate the feasibility of topic maps. Although it is generally not possible and not necessary to say that if a topic map is good enough, we still need a evaluation scheme to measure the feasibility and effectiveness of the topic maps. Here the feasibility concerns about the quality of the retrieved result during the navigation process. The effectiveness refers to the speed that a user reaches his goal by navigation through a topic map. The text mining process in our method may provide a standard for evaluating automatically constructed or even manually constructed topic maps. Further research will be devoted to this subject. Finally, our work may help in semi-automatic topic map authoring. There is a general need for topic map authoring when dealing with large corpora and our method may be the first step in the authoring process. The topics and their occurrences as well as the topic associations all can be easily identified. After such initial step, the authors may further refine the partly constructed topic map by assigning the topic types, occurrence types, occurrence roles, association

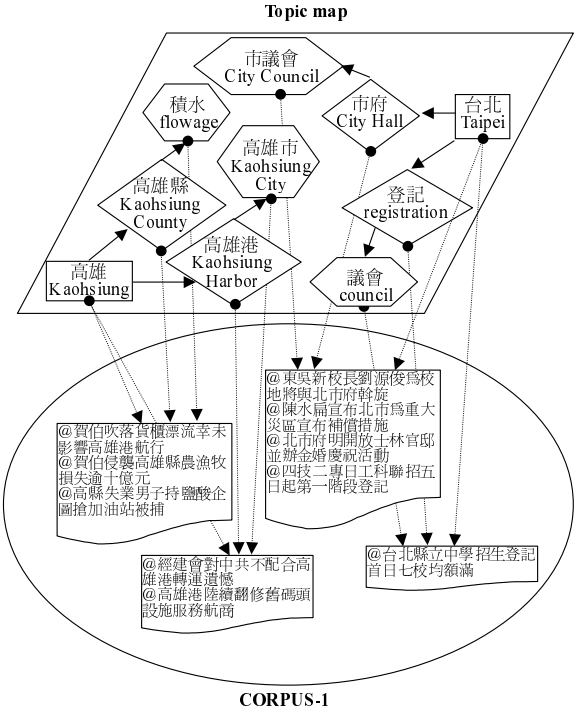


Fig. 4. Part of the topic map of CORPUS-1

types, or even the scopes. We believe that such help will achieve the best result when the fully automatic schemes are still not perfectly developed.

References

1. R. Baeza-Yates and B. Ribeiro-Neto. *Modern Information Retrieval*. Addison Wesley, Reading, MA, 1999.
2. M. Biezunski. Introduction to topic mapping. In *SGML Europe 1997 GCA Conference*, Barcelona, Spain, 1997.
3. C. Fellbaum. *WordNet: An Electronic Lexical Database*. MIT Press, Cambridge, MA, 1998.
4. International Organization for Standardization. *ISO/IEC 13250, Information technology V SGML Applications V Topic Maps*. ISO, Geneva, 2000.
5. T. Kohonen. *Self-Organizing Maps*. Springer-Verlag, Berlin, 1997.
6. C. H. Lee and H. C. Yang. A web text mining approach based on self-organizing map. In *Proc. ACM CIKM'99 2nd Workshop on Web Information and Data Management*, pages 59–62, Kansas City, MI, 1999.
7. G. Moore. Topic map technology - the state of the art. In *XML Europe 2000*, Paris, Frace, 2000.
8. S. Pepper. Navigating haystacks, discovering needles. *Markup Languages: Theory and Practice*, 1:41–68, 1999.

9. H. H. Rath. Technical issues on topic maps. In *Proceedings of Metastructures 99 Conference, GCA*, 1999.
10. G. Salton and M. J. McGill. *Introduction to Modern Information Retrieval*. McGraw-Hill, New York, 1983.
11. H. C. Yang and C. H. Lee. Automatic category structure generation and categorization of chinese text documents. In *Proceedings of The Fourth European Conference on Principles and Practice of Knowledge Discovery in Databases(PKDD-2000)*, pages 673–678, Lyon, France, 2000.